Cancer Survival ML Project | Model Performance

- Goal: predict from clinical data and gene expression whether a patient survives at least 5 years after diagnosis. The model is trained on the 325 tumor samples of the cohort.
- Two ways of condensing ~19,000 genes into a usable input are compared: PCA and DESeq2 (see buttons).
- Top row: Neural network, 5-fold cross-validation: mean ± standard deviation across folds. Lower sections: one train/test split, used to show how the model behaves.

AUROC	MACRO F1	ACCURACY
0.80 ±0.067	0.69 ±0.046	0.72 ±0.039
PCA ▲ 0.044	PCA ▲ 0.058	PCA ▲ 0.058
Ranking quality across all cutoffs	Balanced score across both groups	Share of correct predictions

Deseq2 | PCA

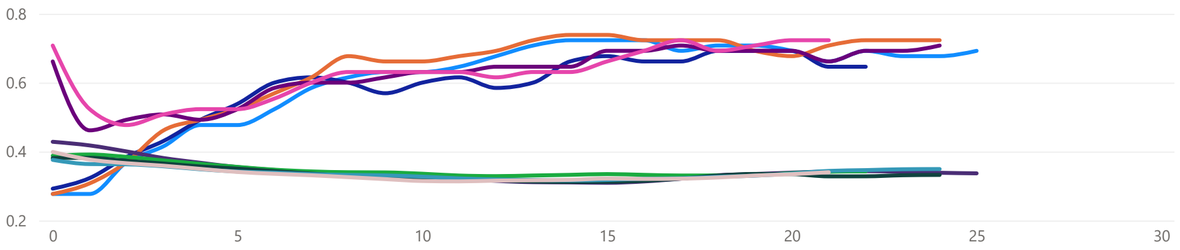
Bookmark toggle - Select the gene-input method (updates the whole page)

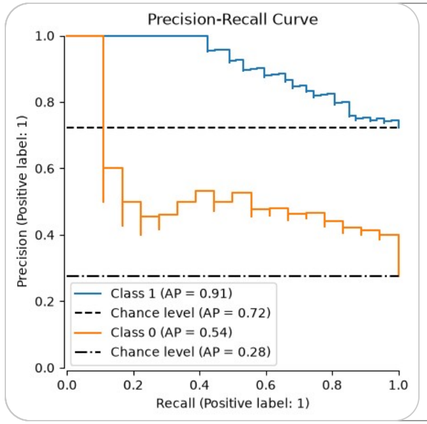
PCA: compresses all genes into 50 summary components that capture most of the variation between patients. It never looks at survival.
DESeq2: keeps only the genes that differ significantly between tumor and normal tissue (adjusted p < 0.05, |log2 fold change| > 2). Chosen by biology, not by survival.

Model behavior - Single Train/Test split

Learning curves: accuracy (upper lines) and loss (lower lines) by training epoch

Each line is one of 5 training runs. Accuracy rises and levels off while loss falls and flattens; training stops early if the model stop improving.





- **Precision** defines how often a patient flagged by the model truly belongs to the positive group.
- **Recall** defines the model capacity to flag the right patients across the whole cohort.
- **AP (average precision)** summarizes the curve, while dashed lines show random guessing. The orange line (Class 0 - non-survivors), are the hardest group to define, due to low number and noisy estimates.

Recall Class 0	Recall Class 1
0.69 ±0.100	0.73 ±0.034
PCA ▲ 0.035	PCA ▲ 0.044

Model comparison - Matching patients cohort (65 samples)

Neural network confusion matrix

Actual	< 5 yrs	> 5+ yrs
< 5 yrs	13	5
> 5+ yrs	15	32

Logistic Regression confusion matrix

Actual	< 5 yrs	> 5+ yrs
< 5 yrs	7	11
> 5+ yrs	6	41

Random Forest confusion matrix

Actual	< 5 yrs	> 5+ yrs
< 5 yrs	1	17
> 5+ yrs	1	46

- **Neural Network:** finds 13 of 18 non-survivor (misses 5); 15 of 47 survivors wrongly flagged; accuracy 69%
- **Logistic Regression:** finds 3 of 18 non-survivors (misses 15); 7 survivors wrongly flagged; predicts "survived" for 55 of 65 patients; accuracy 66%
- **Random Forest:** finds 1 of 18 non-survivors (misses 17); 1 survivor wrongly flagged; predicts "survived" for 63 of 65; accuracy 72%

Overall takeaway:

The neural network ranks patients well (**AUROC 0.85 ± 0.03 with PCA, 0.80 ± 0.07** with DESeq2) and flags **73%** and **69%** of those who died within 5 years. **PCA edges DESeq2**, but only modestly. **Simpler models catch only 1–7 of 18** at-risk patients vs **13–16 for the neural network**, suggesting a need for a model of higher complexity when working with highly dimensional RNA-seq/Metadata.